

Capstone Project Final Presentation – Fairoz Ahmad Sheikh

GitHub

https://github.com/FairozAhmadSheikh/IBM_Assignments

Repository:

Presentation

https://github.com/FairozAhmadSheikh/IBM_Assignments/blob/main/SpaceX-Falcon-9-Launch-Data-Analysis.pdf

PDF:

Executive

Summary:

This capstone project involved a comprehensive exploration of SpaceX launch data, leveraging Python, SQL, and interactive visualization tools. The project workflow included data collection, wrangling, exploratory data analysis, predictive modeling, and dashboard creation. The goal was to gain actionable insights from historical launch data, evaluate launch success factors, and visualize findings effectively.

Introduction:

The project focuses on analyzing SpaceX launch missions, including Falcon 9 launches, payload data, launch sites, mission outcomes, and trends over time. Key objectives included understanding launch success rates, orbit types, mission performance, and applying predictive models to classify launch outcomes.

Data Collection and Wrangling Methodology:

- Retrieved SpaceX launch data via REST API using GET requests.
- Filtered for Falcon 9 launches and cleaned missing values in PayloadMass using mean imputation.
- Standardized categorical columns and prepared the data for analysis.

EDA and Interactive Visual Analytics Methodology:

- Explored launch success trends over years using Matplotlib and Seaborn.
- Calculated launch counts per site and payload statistics.
- Visualized outcomes per orbit type and landing outcomes using bar charts, catplots, and scatterplots.
- Created interactive maps using Folium to display launch sites and landing locations.

Predictive Analysis Methodology:

- Built classification models (Decision Tree, K-Nearest Neighbors, SVM) to predict launch success.
- Evaluated models using train/test splits, cross-validation, and performance metrics like accuracy, F1-score, and confusion matrices.
- Tuned hyperparameters to optimize predictive accuracy.

EDA with Visualization Results:

- Showed yearly launch trends, launch success rates, and payload mass distributions.
- Visualized mission outcomes by orbit type and launch site using Seaborn plots.
- Created scatterplots and categorical plots to highlight patterns in the data.

EDA with SQL Results:

- Loaded SpaceX dataset into SQLite and performed SQL queries for:
 - Launch counts per site
 - Payload mass statistics
 - Mission outcome counts
 - Aggregated payload mass by booster version
- Displayed results using SQL queries with LIMIT, MIN, SUM, and GROUP BY clauses.

Interactive Map with Folium Results:

- Added launch and landing markers using `add_child()` and `add_to()` methods.
- Grouped multiple markers with MarkerCluster plugin.
- Visualized launch success and landing outcomes on a world map.

Plotly Dash Dashboard Results:

- Developed an interactive dashboard to explore launch success, payload, and launch site trends.
- Incorporated drop-downs and sliders to filter data dynamically.
- Connected callbacks to update figures based on user input.

Predictive Analysis (Classification) Results:

- Decision Tree, KNN, and SVM models were evaluated on test data.
- Best accuracy achieved: **83.33%** on the test set.
- Used RBF kernel for SVM, optimized hyperparameters via validation sets.
- Provided insights into factors influencing launch success.

Conclusion:

- Falcon 9 launches show an overall success rate of **67%**, with CCAFS SLC 40 as the most frequently used site.
- Payload mass and launch site are significant factors in launch outcomes.
- Interactive visualizations and dashboards effectively communicate data insights.
- Predictive models can support operational decision-making for future launches.

Creativity and Insights:

- Used Folium maps and Plotly Dash dashboards for enhanced interactivity.
- Derived insights such as trends in drone ship landings and orbit-specific mission outcomes.
- Applied predictive analytics to classify potential launch outcomes, aiding strategic planning.